

Participatory Evolution of Artificial Life Systems via Semantic Feedback

Shuowen Li*
Tsinghua University
Beijing, China
Artineer LLC
Massachusetts, USA
lsw23@mails.tsinghua.edu.cn

Danqi Huang
Tsinghua University
Beijing, China
hdq24@mails.tsinghua.edu.cn

Kexin Wang*[†]
MIT CSAIL
Massachusetts, USA
Artineer LLC
Massachusetts, USA
kexinw@mit.edu

Ali Asadipour
Royal College of Art
London, United Kingdom
ali.asadipour@rca.ac.uk

Yitong Sun[‡]
Royal College of Art
London, United Kingdom
yitong.sun@network.rca.ac.uk

Minglu Fang
China Academy of Art
Hang Zhou, China
2221305075@caa.edu.cn

Haipeng Mi[‡]
Tsinghua University
Beijing, China
mhp@tsinghua.edu.cn

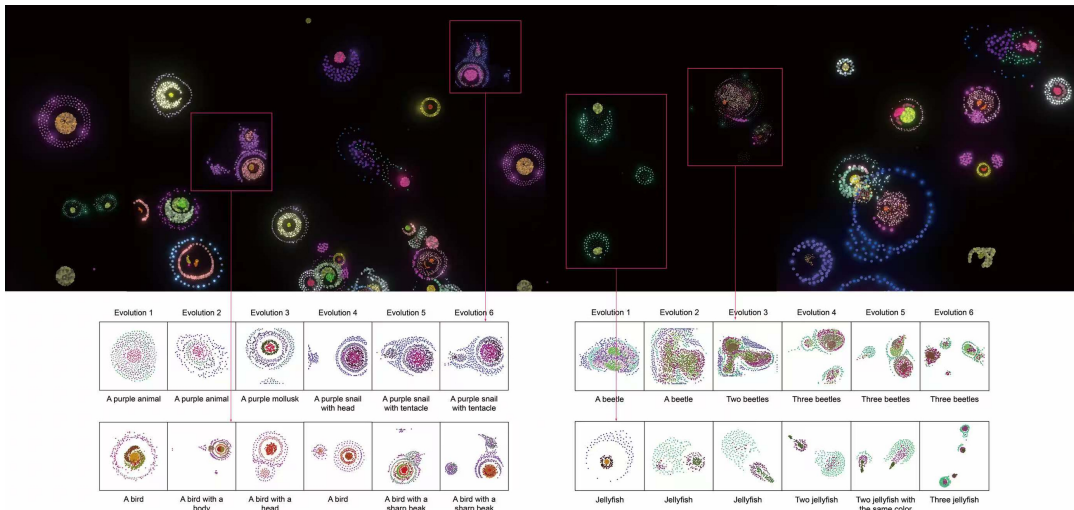


Figure 1: Evolutionary Pathways of Digital Lifeforms and Schematic Illustration of Ecological States in a Digital Life System.

Abstract

We present a semantic-feedback framework that treats natural language as a regulatory signal for evolving artificial-life systems. Instead of using prompts to select finished images, text in our system shapes the dynamics of an interactive ecosystem, allowing

audiences to cultivate behaviors over time. The framework couples a learned mapping from prompts to simulation parameters with evolutionary search and vision-language evaluation, so user intent modulates both visible outcomes and the underlying generative rules. It supports iterative prompt refinement, multi-agent interaction, and the synthesis of new collective rules from community input. In a user study, participants achieved higher semantic alignment and reported a greater sense of control than with manual tuning, while behaviors remained diverse across generations. As an art-led contribution, the work reframes authoring as participatory cultivation and advances open-ended evolution as a socially distributed, not solely algorithmic, process; as a tool contribution, it offers a practical platform for co-creative generative design.

*Both authors contributed equally to this research.

[†]Project Lead

[‡]Corresponding author.



CCS Concepts

• **Applied computing** → **Media arts**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

artificial life; semantic feedback; natural language interaction; generative systems; agent-based simulation.

ACM Reference Format:

Shuowen Li, Kexin Wang, Minglu Fang, Danqi Huang, Ali Asadipour, Haipeng Mi, and Yitong Sun. 2025. Participatory Evolution of Artificial Life Systems via Semantic Feedback. In *SIGGRAPH Asia 2025 Art Papers (SA Art Papers '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3757369.3767620>

1 Introduction

Human interaction with complex systems, including ecological environments, social dynamics, and artificial simulations, presents ongoing challenges for both modeling and creative intervention. While agent-based models and artificial life systems provide formal tools for simulating emergent behavior, they often rely on fixed rule structures and limited input modalities. This makes it difficult to embed high-level conceptual intent into evolving system dynamics.

Recent advances in multimodal AI, particularly vision-language models such as Contrastive Language-Image Pretraining (CLIP), have expanded the potential of using natural language to guide generative systems. However, most implementations adopt a static prompt-to-output paradigm in which language acts as a trigger rather than a continuous modulator. This restricts semantic iteration and reduces a system's ability to adapt behavior in response to evolving user input.

In this work we return to artificial life to make generative processes visible and shapeable by people. Generative tools are now common, yet they often hide how results are produced and why certain behaviors appear. From a critical perspective rather than a purely technical one, the focus is to keep the making in view and to invite negotiation of meaning, not only the selection of final outputs. We therefore treat language as a medium that regulates behavior over time rather than a trigger for one-off results. Our system lets audiences grow, steer, and revise an evolving ecosystem, so authorship becomes shared and procedures remain legible. The focus shifts from matching a prompt to cultivating dynamics that can be discussed and adjusted. In cultural settings, this reframes spectatorship as participation and relocates meaning-making into the ongoing regulation of behavior. We understand open-ended evolution not as endless novelty but as the practice of maintaining diversity, legibility, and revision in a public setting.

Motivated by this view, we introduce a semantic feedback-driven framework for evolving artificial life systems, where natural language serves as both expressive input and regulatory signal. Grounded in a swarm-based model, our architecture integrates a Bidirectional Encoder Representations from Transformers (BERT) prompt-to-parameter encoder, a Covariance Matrix Adaptation Evolution Strategy (CMA-ES) optimizer, and a CLIP-based semantic evaluation module (e.g., [Devlin et al. 2019; Hansen and Kern 2004a; Radford et al. 2021]). This closed-loop pipeline allows language to

influence both the visual output and the behavioral rules of the simulation, enabling interpretable, adaptive, and semantically aligned evolution.

To demonstrate the framework's capabilities, we implemented it in an interactive platform that enables users to co-construct a semantic ecosystem. Participants generate and evolve synthetic agents using language prompts. These agents coexist and interact in a shared artificial environment, exhibiting behaviors shaped by both local rules and emergent collective dynamics. As prompts accumulate, the system analyzes linguistic histories and behavioral trajectories to derive meta-rules that recursively influence subsequent evolution. *A video demonstration is available*¹.

We evaluated the system in two settings: a controlled study assessing its alignment with user intent, and a public deployment supporting open-ended co-creation. Results suggest that semantic input enhances controllability, interpretability, and user engagement, positioning the framework as a novel platform for participatory artificial life and expressive complex system design.

2 Related Work

2.1 Consciousness and the Emergence of Artificial Life

Contemporary theories of consciousness from relational, ecological, and generative perspectives challenge the view that consciousness is uniquely human and a priori. Frameworks such as animism [Harvey 2005], the extended mind [Clark and Chalmers 1998], and embodied cognition [Varela Francisco et al. 1991] re-conceptualize it as a regulatory process arising from system and environment coupling. In this view, consciousness functions as feedback that enables self-organization, boundary modulation, and adaptive responses to perturbations, inspiring artificial systems that embed similar self-regulatory and emergent dynamics.

Agent-based models in the artificial life tradition, including Reynolds' Boids [Reynolds 1987], cellular automata, and swarm-based ecologies [Sayama 2009], offer formal tools to study emergent complexity through decentralized local interaction. These systems show how simple individual rules yield lifelike collective phenomena. Although widely used in generative art, ecological simulation, and swarm robotics, most rely on predefined rules and lack semantically adaptive input, which limits responsiveness to human conceptual intent [Park et al. 2023; Wang et al. 2024].

This motivates language-integrated artificial life systems. Rather than treating language as a symbolic description of output, we use it as continuous regulatory input analogous to consciousness, shaping system evolution in real time. Our work positions semantic feedback as a mediator between user intent and behavioral adaptation, extending artificial life through participatory, linguistically guided evolution.

2.2 Language-guided generative systems and semantic feedback

Natural language is a common interface to generative systems, but it is usually a static descriptor of desired outputs. Vision-language models such as Contrastive Language-Image Pretraining (CLIP)

¹Supplementary video: <https://youtu.be/KStCmXm1luw>

learn aligned embeddings used to rank or guide images relative to text [Radford et al. 2021]. Large language models and encoders such as Bidirectional Encoder Representations from Transformers (BERT) provide text embeddings that can be mapped to control spaces or parameter vectors [Devlin et al. 2019]. Derivative-free evolutionary strategies such as Covariance Matrix Adaptation Evolution Strategy (CMA-ES) effectively explore high-dimensional, non-convex objectives without gradients [Hansen and Kern 2004b].

Most language-integrated pipelines adopt a prompt-to-output paradigm: text specifies a target, the system produces a single result, and the process restarts with a new prompt. Even when CLIP is used in an optimization loop, language often acts as a fixed objective rather than a temporally sustained regulator of behavior. Work on language-mediated agents, for example Generative Agents [Park et al. 2023], shows how text, memory, and planning can script social behaviors in simulated worlds, but control is symbolic or discrete and does not continuously modulate the low-level rules that generate physical or swarm-like dynamics.

Our approach differs along three axes of semantic regulation. First, we treat language as a continuous feedback signal that persists during runtime; prompts can be refined and their influence accumulates through time. Second, regulation occurs at the level of behavior-generating rules: text steers parameters of an artificial-life simulation, and evaluation in a vision-language space closes the loop by measuring semantic alignment of evolving behaviors [Radford et al. 2021]. Third, the framework supports participatory meta-dynamics: aggregated linguistic histories and behavioral trajectories are analyzed to synthesize meta-rules that recursively shape subsequent evolution.

By combining a BERT-based mapping from prompts to parameter spaces [Devlin et al. 2019] with CMA-ES for exploration [Hansen and Kern 2004b] and CLIP for semantic evaluation [Radford et al. 2021], the system reframes language from description to regulation. This positions our work at the intersection of artificial life, vision-language alignment, and evolutionary search, while addressing limitations of static prompt pipelines by enabling interpretable, adaptive, and socially negotiated control over emergent dynamics.

2.3 Participatory Artificial Ecosystems and Generative Art

Artificial life has long intersected with generative art through systems that embed user input into emergent behavioral frameworks. Pioneering works such as Sommerer and Mignonneau’s Interactive Plant Growing (1994) and A-Volve (1995) [Sommerer and Mignonneau 1997, 2002] enabled physical engagement with biological or digital organisms, where touch and gesture drove growth and transformation. Other projects extended human-guided evolution: Karl Sims’s Galápagos (1997) [Cárdenas et al. 2014] let gallery visitors select preferred individuals as a fitness proxy, and TechnoSphere (1995–2002) [Prophet 2001] let users specify morphology, movement style, and behavioral traits for creatures released into a shared world that reproduced, competed for resources, and underwent ecosystem-level selection.

While these systems show how user input can shape artificial evolution, interactions are typically limited to gestures, parameter choices, or predefined symbolic controls. Building on this lineage,

our framework introduces natural language as a direct and expressive channel for intent. We propose a closed-loop architecture in which linguistic input continuously modulates a particle-based artificial life simulation. The system evolves through agent-driven structural emergence and ongoing semantic alignment between user prompts and generative behavior. We implement this framework in a publicly accessible interactive platform, demonstrating its potential as a participatory system for language-driven artificial ecosystems and generative art.

3 Methodology

Unlike image generation tasks, which typically operate under well-defined objectives and constrained solution spaces, the optimization of complex systems presents fundamental challenges, including high-dimensional coupling, nonlinearity, and emergent dynamics. Agent-based simulations such as Swarm exemplify these properties, as they feature intricate dependencies among agents, local interaction rules, and environmental variables, without any explicit or globally optimal state.

Although evolutionary algorithms such as CMA-ES and genetic algorithms have been extensively used in these black-box contexts, they rely heavily on stochastic sampling and are often susceptible to local optima, especially in high-dimensional search spaces [Uchida et al. 2024]. Recent advances in reinforcement learning have introduced dynamic control strategies [Gupta et al. 2020], yet few methods integrate real-time semantic feedback or support conceptual intent as a guiding force for system evolution in a flexible and interpretable manner.

To address this gap, we propose a framework for semantically guided optimization of artificial life systems via natural language input. Grounded in the Swarm model, our approach establishes a closed-loop pipeline that links semantic prompts to behavioral rule search and iterative evolutionary refinement. This framework replaces undirected exploration with quasi-random search shaped by linguistic guidance [Kumar et al. 2024], leveraging the expressive capacity of natural language to introduce meaningful directionality without relying on differentiable objectives.

By treating semantic intent as an active regulatory mechanism within an agent-based environment, our approach facilitates intuitive, high-level intervention in complex system behavior. This reduces the technical barrier to engaging with artificial life simulations and opens new possibilities for collaborative human-machine experimentation, particularly in generative design, interactive art, and interdisciplinary research.

3.1 Closed-Loop Semantic Feedback Architecture

As shown in Figure 2, the system comprises three core components:

- **Language Encoding (Prompt2Param):** Natural language prompts are translated into parameter vectors that encode semantic intent.
- **Parameter Optimization (CMA-ES):** Based on these vectors, the system performs evolutionary search to optimize agent-based behavioral configurations.

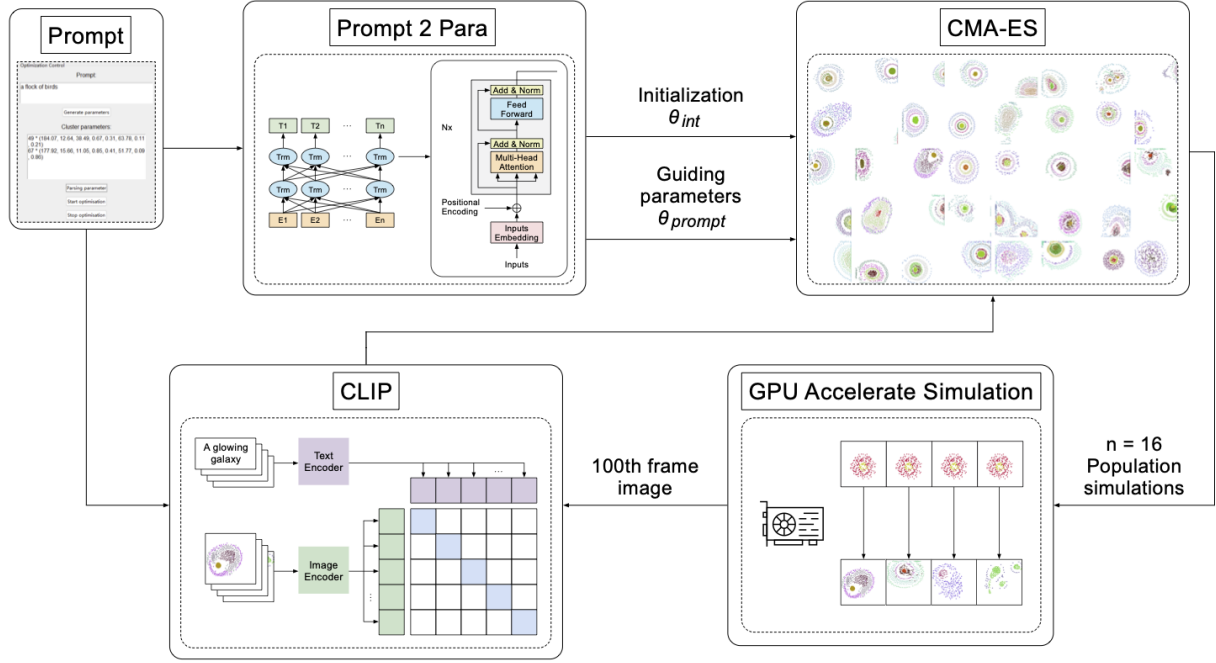


Figure 2: System architecture of the semantic feedback driven evolution framework. A natural language prompt is encoded into parameters that steer a GPU-accelerated, population-based simulation. CMA-ES evaluates populations via CLIP’s transformer-based image and text embeddings, computing semantic-similarity fitness between the prompt and simulation outputs and iteratively updating parameters to align behavior with user intent, yielding real-time visual feedback across generations.

- **Semantic Evaluation (CLIP):** A vision-language model quantifies the semantic similarity between simulation outputs and the original prompt, supplying feedback to guide the optimization process.

Natural language lets users express abstract, affective, or metaphorical goals (e.g., “expand like a nebula,” “gather like magnets”) that are hard to specify numerically. To connect such high-level expressions to low-level configurations, we built a dataset of 100 prompts paired with Swarm parameters and trained Prompt2Param, a lightweight BERT encoder, to output two vectors: an initialization vector θ_{init} for starting the simulation and a semantic bias θ_{prompt} that acts as a directional prior during optimization. This dual design provides stable initialization and sustained semantic guidance across generations.

For search we use CMA-ES, well suited to high dimensional, non convex, non differentiable problems. Unlike gradient methods, CMA-ES works in black box settings and requires no explicit loss. Its modularity lets us inject language derived priors without compromising stability or convergence.

Alignment is assessed with a CLIP based cross modal fitness. Each simulation yields a frame sequence; representative images are encoded with CLIP ViT-B/32 into 512 dimensional embeddings and compared with the prompt embedding via cosine similarity to produce a semantic fitness score. This concept driven signal replaces

hand crafted visual metrics, steering evolution toward behaviors that match human descriptions rather than rigid templates.

3.2 Semantic-Driven Evolutionary Workflow

When a user submits a natural language prompt, the system initiates a semantically guided optimization process. The Prompt2Param module encodes the prompt into two output vectors, the vector θ_{init} defines the initial behavioral configuration of the Swarm simulation, including parameters such as speed, neighborhood radius, and cohesion strength. Meanwhile, θ_{prompt} functions as a directional prior throughout the evolutionary process, guiding the system toward behaviors aligned with the user’s intended semantics.

Starting from θ_{init} , the CMA-ES algorithm samples a population of $n = 16$ candidate parameter sets per generation. Each candidate defines a complete behavioral rule set for the Swarm model, including continuous variables such as neighborhood radius, maximum velocity, alignment coefficient, cohesion coefficient, separation coefficient, and noise intensity.

Each parameter set is used to run a full GPU-accelerated simulation. The resulting particle trajectories are rendered into images, which are then processed through the CLIP ViT-B/32 model to generate 512-dimensional visual embeddings. These embeddings are compared against the text embedding of the original prompt

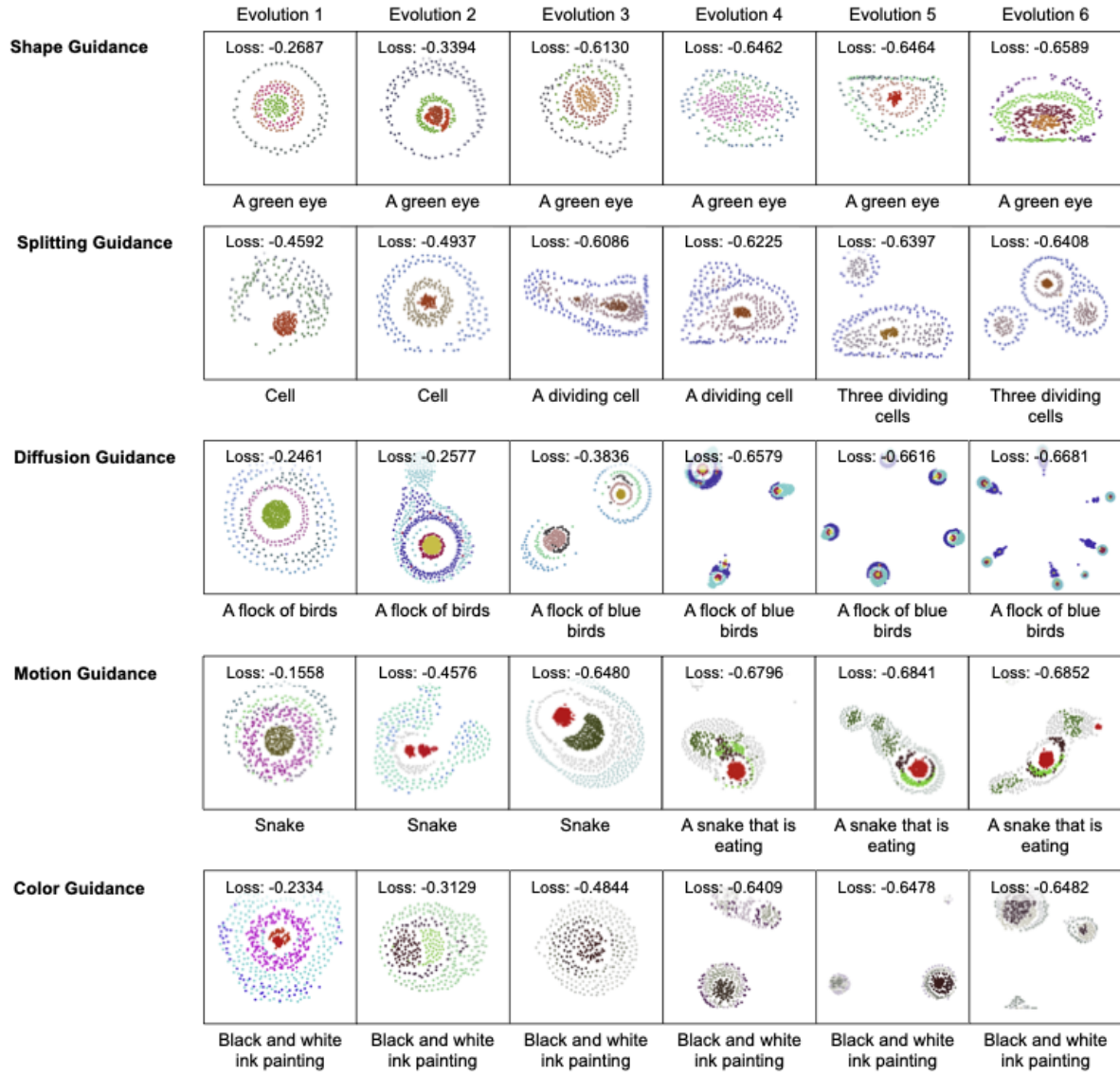


Figure 3: Evolutionary trajectories under five semantic prompts. Each row shows six iterations. From top to bottom: "Black and white ink painting," "A snake that is eating," "Three dividing cells," "A green eye," and "A flock of blue birds." Loss values under each image are CLIP-based semantic distances between prompt and output. Prompts guide behavior in a closed loop, iteratively shaping emergent visual patterns.

using cosine similarity, yielding a semantic fitness score for each candidate.

CMA-ES aggregates the fitness scores across the population to update its sampling distribution, progressively converging toward regions of the parameter space that produce behaviors more semantically aligned with the user's intent. Throughout this process,

θ_{prompt} continues to act as a guiding vector, biasing the search in a conceptually coherent direction.

To prevent premature convergence and ensure creative variability, the system introduces mild stochastic noise and parameter mutations at each generation. It also monitors the diversity of visual embeddings across generations. This mechanism preserves visual

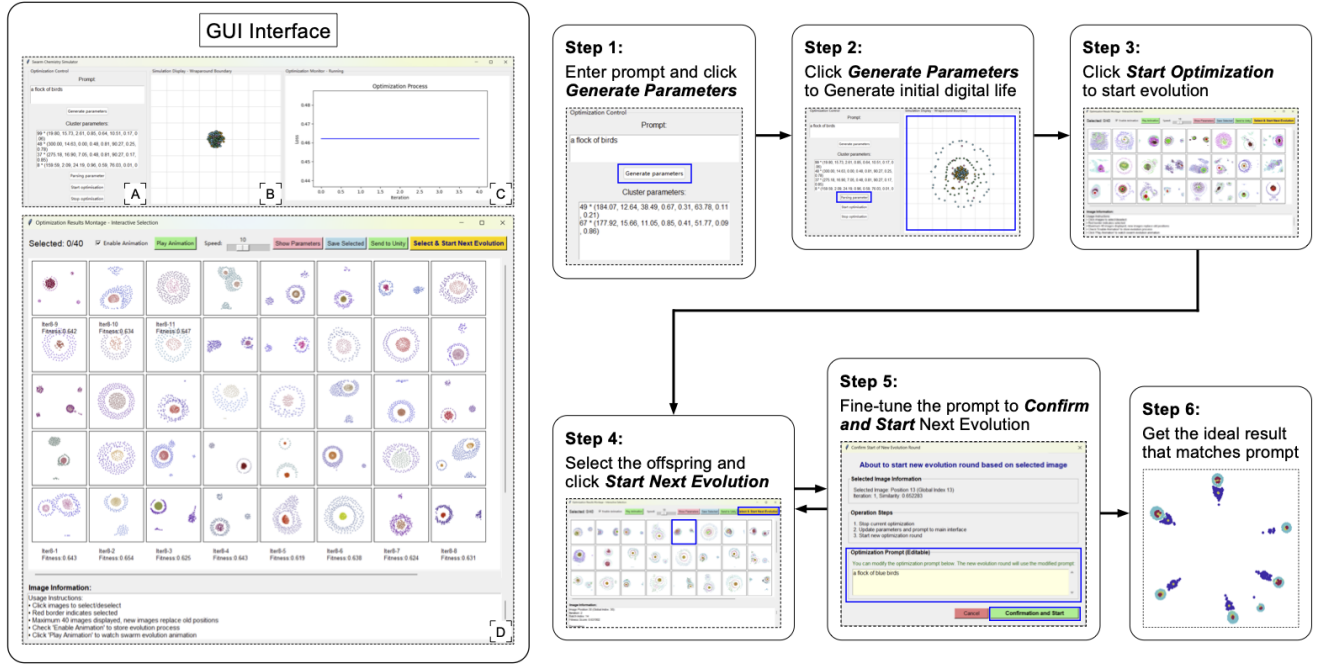


Figure 4: User interaction workflow within our semantic-driven generative system. (A) Users input a prompt and generate the initial parameter configuration. **(B)** The system produces an initial simulation based on this prompt. **(C)** Evolution is initiated through a semantic optimization loop, and users can select preferred outcomes to guide the next generation. **(D)** Users may refine the prompt at any stage to steer the system toward their conceptual intent. The process continues until the emergent behavior aligns with the user’s evolving goals.

richness and behavioral diversity, while maintaining semantic consistency with the user-defined prompt. Figure 3 presents examples of prompt-guided evolution, visually demonstrating how the system refines agent behavior across generations under different semantic conditions.

3.3 System GUI Integration

To enhance accessibility and ease of use, we developed a graphical user interface (GUI) designed for artists, designers, and other users without a programming background. Built with Python and Tkinter, the tool allows intuitive interaction with the semantic evolution of complex systems.

Figure 4 illustrates the interface layout. Users begin by entering a natural language prompt into the interface. With a single click, the system initiates the full optimization process: the prompt is encoded into simulation parameters, the CMA-ES algorithm performs evolutionary search, and the CLIP model evaluates the semantic alignment of each generation. The interface presents real-time visualization of the Swarm simulation results, enabling users to monitor evolutionary progress, select preferred outcomes, or launch new rounds of optimization based on chosen individuals.

4 Application: A Game of Ecosystem Construction

To demonstrate the feasibility of our semantic complex systems framework, we developed an interactive game in which players collaboratively construct and evolve an artificial ecosystem. The experience unfolds in three stages, each reflecting a shift in system dynamics.

Stage 1: Individual Symbolic Construction. Players access the game via a web interface and submit a natural language prompt, which is encoded by the Prompt2Param module into an initial Swarm configuration. This is sent via WebSocket to a Python-based evolution engine running CMA-ES, which returns real-time visualizations to the client. Players can pause evolution, revise prompts, and spawn new branches, iteratively shaping digital lifeforms that reflect their conceptual intent. This stage links symbolic language to agent-based dynamics, allowing abstract ideas to manifest as generative behavior.

Stage 2: Multi-Agent Ecological Interaction. Once a lifeform is finalized, its behavioral parameters are transferred to a shared Unity-based simulation. Leveraging GPU compute shaders, the platform simulates tens of thousands of agents in real time. Lifeforms enter the virtual environment via network transmission and interact

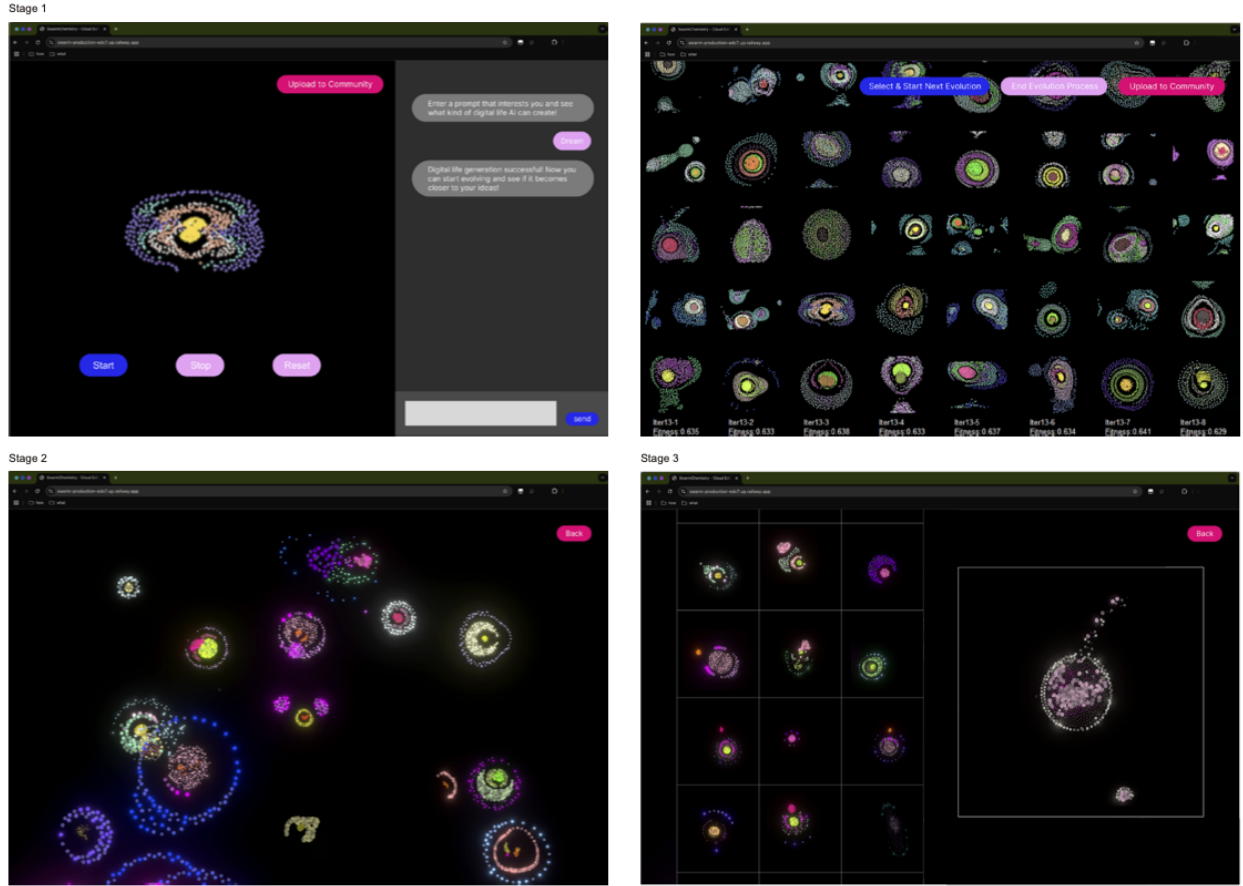


Figure 5: User interface of the participatory semantic-feedback platform across three stages. Stage 1 (top left): single-prompt evolution with live text input and refinement; users can start/stop runs and upload results to the community. Stage 1 – gallery (top right): candidate populations across generations with per-candidate semantic scores; users select a parent to continue evolution or upload outcomes. Stage 2 (bottom left): shared ecosystem where agents evolved from different prompts coexist and interact in real time. Stage 3 (bottom right): community archive and curation view; users browse contributions, inspect behaviors, and select exemplars whose patterns inform meta-rule synthesis re-injected into subsequent runs (see Section 4).

with others, merging, repelling, hybridizing, or asserting territorial boundaries. The simulation is broadcast live through Open Broadcaster Software (OBS) and Restream.io, supporting remote viewing and commentary. These behaviors emerge from both local agent rules and global stochastic effects. User-defined parameters not only shape individual agents but also influence the evolution of the shared environment, embedding conceptual intent into collective dynamics.

Stage 3: Emergent Collective Regulation. As the ecosystem expands, the system records prompt histories, parameter trajectories, and behavioral outcomes. These are clustered and reduced via Principal Component Analysis (PCA) to extract semantic themes, conflict zones, and co-evolutionary patterns. Based on these insights, the system derives emergent “collective rules,” which represent higher-order behavioral logics shaped by dominant or negotiated

tendencies among contributors. These meta-rules are re-injected into the simulation, affecting global dynamics beyond any single agent’s influence. In this final stage, semantic input reorganizes not only individual behavior but the generative logic of the ecosystem itself.

5 Case Studies and User Scenarios

We conducted two complementary user studies to evaluate both the semantic performance and participatory potential of our system.

Controlled Evaluation of Semantic Guidance. The first study assessed the system’s ability to (1) generate agent behaviors aligned with natural language prompts, (2) support directional control through iterative prompt refinement, and (3) provide greater accessibility and expressiveness compared to manual parameter tuning. Ten participants completed three tasks in a controlled lab setting.

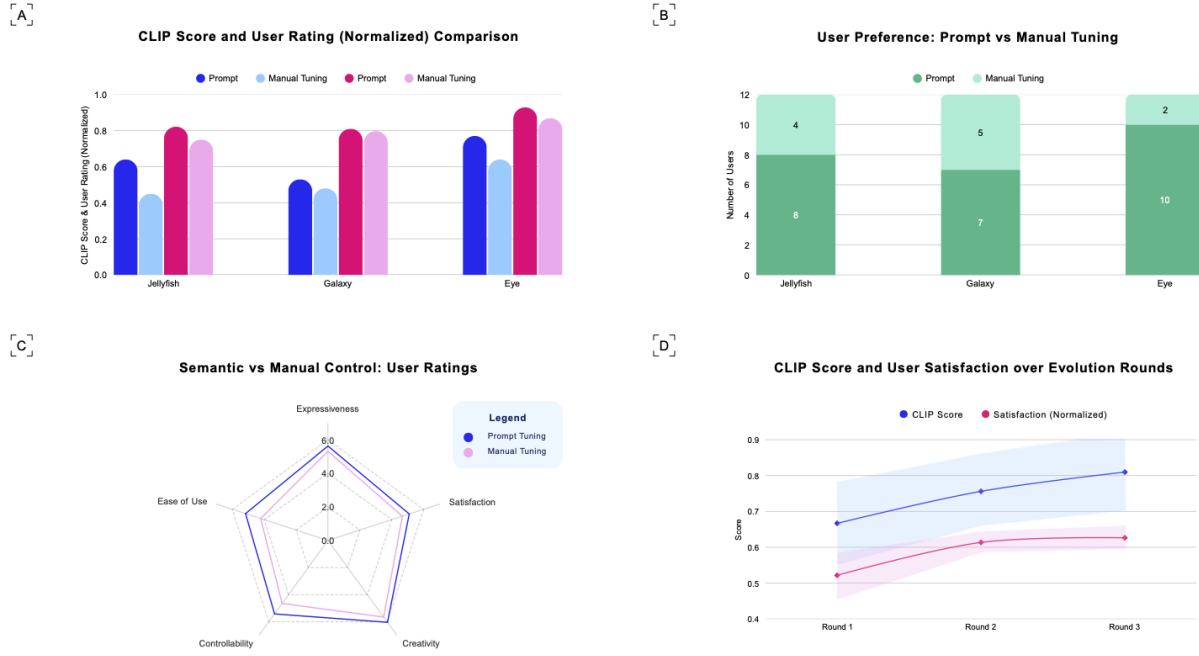


Figure 6: Comparative user evaluation of prompt-based semantic tuning versus manual parameter tuning. (A) CLIP score and normalized user ratings (0–1 scale) across three prompts ("Jellyfish," "Galaxy," "Eye") show consistently higher alignment and satisfaction under prompt-based tuning. (B) User preference counts favor prompt tuning over manual tuning for each prompt. (C) Aggregate user ratings across five dimensions—expressiveness, satisfaction, creativity, controllability, and ease of use—demonstrate a consistent advantage for semantic control. (D) Multi-round analysis of prompt refinement indicates progressive improvement in both CLIP score and satisfaction over three iterations.

In Task A, prompt-driven outputs guided by semantic feedback outperformed manually tuned results across three test prompts, measured by CLIP similarity scores, normalized user ratings, and preference counts (Figure 6A,B). In Task B, participants refined their prompts over three rounds, resulting in progressively higher CLIP alignment and satisfaction ratings (Figure 6C), demonstrating the effectiveness of language as an iterative control modality. Task C compared prompt-based interaction with manual tuning across five user-defined criteria, with language input consistently rated higher in creativity, controllability, and ease of use (Figure 6D). These findings support the system’s capacity for interpretable evolution while enhancing user experience in exploratory design.

Public Deployment in a Generative Ecosystem. To evaluate the system in an open-ended setting, we deployed it in a participatory generative ecosystem game exhibited at the School of Arts and Design, Tsinghua University. This large-scale deployment, described in Section 5, demonstrated the framework’s effectiveness as a real-time creative platform and enabled observations of emergent social dynamics. Participants contributed prompts ranging from poetic abstractions to personal metaphors, with agents cohabiting and interacting in a shared simulation. The evolving ecosystem reflected both individual semantics and collective influence. These outcomes

highlight the framework’s potential to support language-driven co-creation in participatory art and artificial life environments.

6 Discussion

Our findings indicate that semantic feedback is an intuitive, expressive interface for complex generative systems. Compared with manual parameter tuning, natural language conveys abstract goals and metaphors without technical skill. In a controlled study, participants reported higher satisfaction and perceived control with prompt based interaction, and iterative refinement let intent co evolve with system behavior.

Beyond usability, we clarify open endedness supported by three mechanisms: (1) temporal persistence of regulation, where language acts as a sustained signal whose influence accumulates across generations via prompt refinement and history analysis; (2) rule level modulation, where text steers the behavior generating rules of a swarm rather than merely ranking outputs, keeping procedures legible and revisable; and (3) diversity maintenance, where mild stochasticity, parameter mutation, and cross generation embedding monitoring resist premature convergence while preserving semantic alignment (Section 3.2). In public deployment, prompts from multiple contributors informed meta rules synthesized from

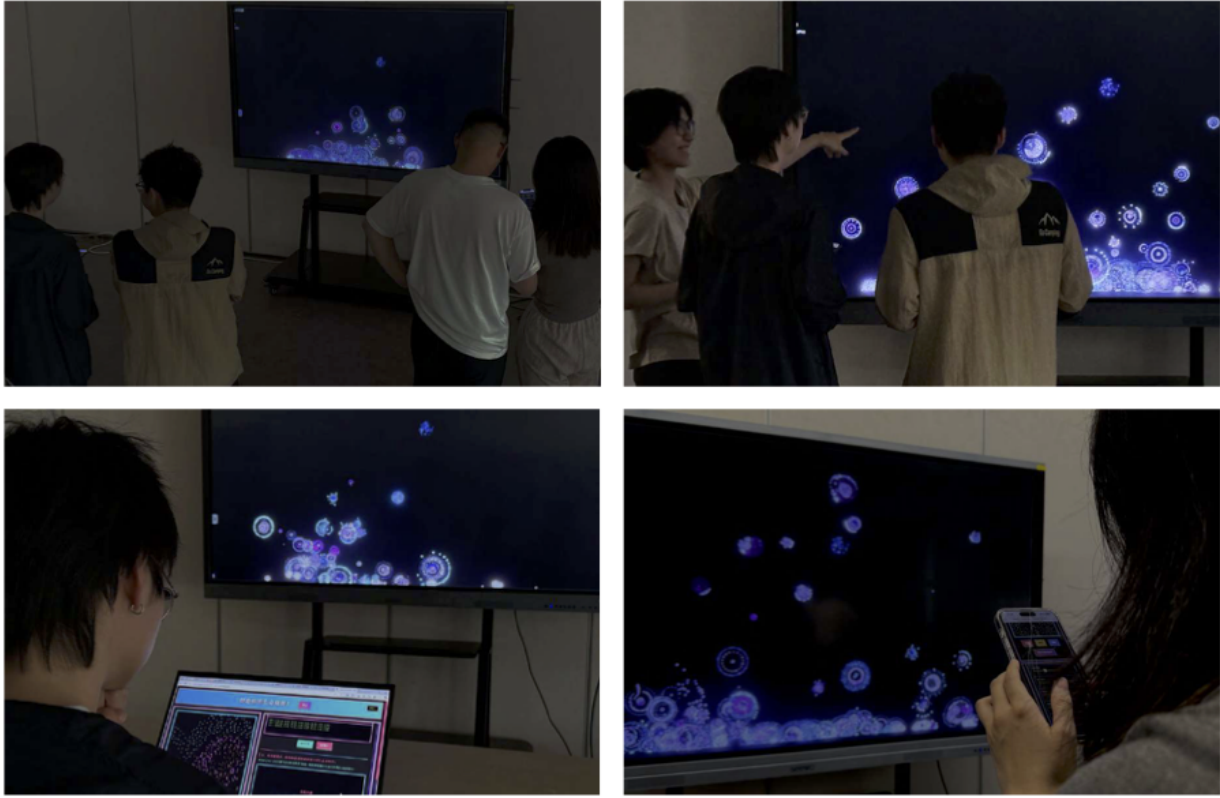


Figure 7: On site public interaction with the language guided artificial life installation. Participants submit natural language prompts on their devices to steer a shared large-screen simulation, co-creating emergent behaviors in real time.

linguistic and behavioral histories, producing ecosystem shifts that reflected collective negotiation.

These mechanisms enable verification of open ended properties: we track cross generation embedding distances among top candidates to monitor variety; compare meta rule injection against ablations without history aggregation; and test whether diversity persists when no new prompts are added for several generations. Together, the evidence suggests that open endedness here is a sustained practice of maintaining diversity, legibility, and revision under semantically meaningful guidance.

7 Conclusion and Limitation

We present a semantic feedback framework in which natural language regulates artificial life via a closed loop of text encoding, evolutionary search, and vision language evaluation. In a controlled study and a public deployment, users achieved higher semantic alignment and reported greater perceived control than with manual tuning, while diversity persisted across generations. Constraints remain: evaluation in a vision language space can reflect cultural and linguistic biases; the prompt to parameter mapping was trained on a small corpus that may miss rare metaphors or niche behaviors;

CLIP based fitness from rendered snapshots can overlook long horizon or non visual qualities; and interactive use depends on GPUs and stable networks, which affects access. To address ethics, we minimize on site data collection, store only non identifying logs, avoid personal imagery, disclose evaluation criteria, allow prompt edits, and encourage plural participation. Overall, the work shifts focus from selecting outputs to cultivating dynamics and offers a practical, participatory medium for open ended co creation and the study of complex systems.

Acknowledgments

We gratefully acknowledge the Complex System Media Lab at Artinneeer LLC (USA) for research support, and the School of Arts and Design at Tsinghua University for providing the exhibition venue for our participatory ecosystem game. Thanks to Haoran Li from The Chinese University of Hong Kong for the technical support on Unity.

References

- L Vaca Cárdenas, DE Olmedo Vizuela, A Tavernise, L Gabriele, F Bertacchini, P Pantano, and Eleonora Bilotta. 2014. Darwin has come back to the Galápagos Islands: An

- educational journey to discover biological evolution. In *EDULEARN14 proceedings*. IATED, 6088–6095.
- Andy Clark and David Chalmers. 1998. The extended mind. *analysis* 58, 1 (1998), 7–19.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *NAACL-HLT*.
- Shubham Gupta, Rishi Hazra, and Ambedkar Dukkipati. 2020. Networked multi-agent reinforcement learning with emergent communication. *arXiv preprint arXiv:2004.02780* (2020).
- Nikolaus Hansen and Stefan Kern. 2004a. Evaluating the CMA evolution strategy on multimodal test functions. In *International conference on parallel problem solving from nature*. Springer, 282–291.
- Nikolaus Hansen and Stefan Kern. 2004b. Reducing the Time Complexity of the Derandomized CMA-ES. *Evolutionary Computation* (2004).
- Graham Harvey. 2005. *Animism: Respecting the living world*. Wakefield Press.
- Akarsh Kumar, Chris Lu, Louis Kirsch, Yujin Tang, Kenneth O Stanley, Phillip Isola, and David Ha. 2024. Automating the Search for Artificial Life with Foundation Models. *arXiv preprint arXiv:2412.17799* (2024).
- Joon Sung Park, Joseph O'Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th annual acm symposium on user interface software and technology*. 1–22.
- Jane Prophet. 2001. TechnoSphere: “Real” time, “artificial” life. *Leonardo* 34, 4 (2001), 309–312.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, et al. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *ICML*.
- Craig W Reynolds. 1987. Flocks, herds and schools: A distributed behavioral model. In *Proceedings of the 14th annual conference on Computer graphics and interactive techniques*. 25–34.
- Hiroki Sayama. 2009. Swarm Chemistry. *Artificial Life* 15, 1 (2009), 105–114. doi:10.1162/artl.2009.Sayama.swarmchem.001
- Christa Sommerer and Laurent Mignonneau. 1997. Interacting with artificial life: A-Volve. *COMPLEXITY-NEW YORK- 2* (1997), 13–22.
- Christa Sommerer and Laurent Mignonneau. 2002. A-volve. *Leonardo* 35, 5 (2002), 481–481.
- Kento Uchida, Teppei Yamaguchi, and Shinichi Shirakawa. 2024. Covariance Matrix Adaptation Evolution Strategy for Low Effective Dimensionality. *arXiv preprint arXiv:2412.01156* (2024).
- J Varela Francisco, Thompson Evan, and Rosch Eleanor. 1991. The embodied mind: Cognitive science and human experience.
- Kexin Wang, Ivy He, Jinke Li, Ali Asadipour, and Yitong Sun. 2024. Exploring Fungal Morphology Simulation and Dynamic Light Containment from a Graphics Generation Perspective. In *SIGGRAPH Asia 2024 Art Papers (SA '24)*. Association for Computing Machinery, New York, NY, USA, Article 15, 8 pages. doi:10.1145/3680530.3695440